Conclusion

In conclusion, the OptiCrop system successfully demonstrates the application of machine learning techniques in the domain of smart agriculture for accurate crop recommendation. The system effectively utilizes key soil and environmental parameters, including nitrogen, phosphorus, potassium (NPK) levels, temperature, humidity, soil pH, and rainfall, to predict the most suitable crop for a given set of conditions.

The integration of multiple machine learning algorithms such as K-Nearest Neighbors (KNN), K-Means Clustering, and Logistic Regression enhances the robustness and reliability of the prediction system. Each algorithm contributes distinct advantages—ranging from pattern identification and clustering of similar environmental conditions to precise classification of crop suitability—thereby improving overall model performance and generalization capability.

The comparative use of these models ensures higher accuracy and strengthens decision-making support for agricultural applications. The system effectively reduces uncertainty in crop selection and enables farmers to make informed, data-driven decisions.

Overall, OptiCrop highlights the potential of artificial intelligence in transforming traditional agricultural practices into efficient, predictive, and sustainable farming systems.